United States Patent Application Publication	20250283432
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	Ross; Steven A. et al.

FUEL INJECTOR COOLING SYSTEM

Abstract

A fuel injector cooling system, comprising: a blower including a blower inlet and a blower outlet; a valve including a valve inlet and a valve outlet; a duct defining a flow passage between the blower outlet and the valve inlet; a fuel injector including a flange and a valve housing extending radially outward from the flange, wherein the valve housing defines an outer surface of the fuel injector; and a cooling jacket defining an inner surface and a cooling air inlet in fluid communication with the flow passage, wherein the cooling jacket defines a cooling flow passage between the inner surface of the cooling jacket and the outer surface of the valve housing, wherein the cooling flow passage is in fluid communication with the flow passage.

Inventors: **Ross; Steven A. (Wyoming, OH), Laubenthal; Jeff Scott (Warrendale, PA), Stevens; Donald Craig (Amelia, OH), Kress; Eric John (Loveland, OH)**

Applicant: **General Electric Company** (Schenectady, NY)

Family ID: **96948712**

Appl. No.: **18/599505**

Filed: **March 08, 2024**

Publication Classification

Int. Cl.: **F02C7/232** (20060101); **F02C7/18** (20060101)

U.S. Cl.:

CPC **F02C7/232** (20130101); **F02C7/18** (20130101); F05D2240/35 (20130101); F05D2260/232 (20130101)

Background/Summary

FIELD

[0001] The present disclosure relates to gas turbine engines. More particularly, this disclosure is directed to fuel injector cooling systems for a gas turbine engine.

BACKGROUND

[0002] A gas turbine engine generally includes a fuel system including various valves which provide a supply of hydrocarbon fuel (liquid or gas) to fuel injectors of a combustor during operation. When the gas turbine engine is shut down, engine cooling systems may also shut down, resulting in thermal soakback where residual heat in certain engine components is transferred to other components such as the fuel valves as the engine cools. As the fuel valves heat up, carbon deposits or "coke" may form in the fuel valves.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] A full and enabling disclosure of the present disclosure, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures, in which:

[0004] FIG. **1** is a perspective view of an exemplary aircraft in accordance with an exemplary embodiment of the present disclosure.

[0005] FIG. **2** is a cross-sectional schematic view of a gas turbine engine in accordance with an exemplary aspect of the present disclosure.

[0006] FIG. **3** is a perspective view of an exemplary fuel injector according to an exemplary embodiment of the present disclosure.

[0007] FIG. **4** is an enlarged cross-sectional schematic view of a portion of the gas turbine engine as shown in FIG. **2**, including a portion of a core engine and a fuel injector cooling system according to exemplary embodiments of the present disclosure.

[0008] FIG. **5** is a perspective view of an exemplary fuel injector including a cooling jacket surrounding a valve housing of the fuel injector according to an exemplary embodiment of the present disclosure.

[0009] FIG. **6** is a cutaway view of a top portion of the fuel injector shown in FIG. **5** including the cooling jacket, according to exemplary embodiments of the present disclosure.

[0010] FIG. **7** is an exploded perspective view of an exemplary cooling jacket according to exemplary embodiments of the present disclosure.

[0011] FIG. **8** is an enlarged view of a portion of the cooling jacket and the fuel injector as shown in FIG. **5**, according to exemplary embodiments of the present disclosure.

[0012] FIG. **9** is a perspective view of a portion of the fuel injector cooling system as shown in FIG. **4**, according to an exemplary embodiment of the present disclosure.

DETAILED DESCRIPTION

[0013] Reference will now be made in detail to present embodiments of the disclosure, one or more examples of which are illustrated in the accompanying drawings. The detailed description uses numerical and letter designations to refer to features in the drawings. Like or similar designations in the drawings and description have been used to refer to like or similar parts of the disclosure.

[0014] The word "exemplary" is used herein to mean "serving as an example, instance, or illustration." Any implementation described herein as "exemplary" is not necessarily to be construed as preferred or advantageous over other implementations.

[0015] As used herein, the terms "first," "second," and "third" may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components.

[0016] The terms "forward" and "aft" refer to relative positions within a gas turbine engine or vehicle and refer to the normal operational attitude of the gas turbine engine or vehicle. For example, regarding a gas turbine engine, forward refers to a position closer to an engine inlet section and aft refers to a position closer to an engine nozzle or exhaust.

[0017] The terms "upstream" and "downstream" refer to the relative direction with respect to fluid flow in a fluid pathway. For example, "upstream" refers to the direction from which the fluid flows, and "downstream" refers to the direction to which the fluid flows.

[0018] The terms "coupled," "fixed," "attached to," and the like refer to both direct coupling, fixing, or attaching, as well as indirect coupling, fixing, or attaching through one or more intermediate components or features, unless otherwise specified herein.

[0019] The singular forms "a," "an," and "the" include plural references unless the context clearly dictates otherwise.

[0020] Approximating language, as used herein throughout the specification and claims, is applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as "about," "approximately," and "substantially," are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. For example, the approximating language may refer to being within a 10 percent margin.

[0021] The present disclosure is generally related to a fuel injector cooling system for a fuel injector of a gas turbine engine. More specifically, the disclosure is directed to a cooling system including a blower, a valve downstream from the blower, a duct defining a flow passage between the blower and the valve, a fuel injector including a flange and a valve housing extending radially outward from the flange. The fuel injector cooling system further includes a cooling jacket that at least partially surrounds the valve housing and defines a cooling flow passage between an inner surface of the cooling jacket and an outer surface of the valve housing. In alternate embodiments, the cooling jacket or more particularly, the cooling passage, may be integrally formed, such as via additive manufacturing or other known manufacturing processes, with the valve housing.

[0022] During normal operations of a gas turbine engine, temperatures of the various engine components are maintained within allowable limits by a plurality of cooling systems that transfer heat from the various engine components to one or more heat sinks. When the engine is shutdown, most cooling systems no longer operate. Residual heat in certain engine components (e.g. "soakback") can increase the temperature of other engine components beyond allowable limits. A particular concern is the formation of carbon (or "coke") deposits in fuel carrying components, particularly a fuel injector, when a hydrocarbon fuel (liquid or gas) is exposed to high temperatures in the presence of oxygen.

[0023] Referring now to the drawings, wherein identical numerals indicate the same elements throughout the figures, FIG. **1** is a perspective view of an exemplary aircraft **10** that may incorporate at least one exemplary embodiment of the present disclosure. As shown in FIG. **1**, the aircraft **10** has a fuselage **12**, wings **14** attached to the fuselage **12**, and an empennage **16**. The aircraft **10** further includes a propulsion system **18** that produces a propulsive thrust to propel the aircraft **10** in flight, during taxiing operations, etc. Although the propulsion system **18** is shown attached to the wing(s) **14**, in other embodiments it may additionally or alternatively include one or more aspects coupled to other parts of the aircraft **10**, such as, for example, the empennage **16**, the fuselage **12**, or both.

[0024] The propulsion system **18** includes at least one turbomachine. In the exemplary embodiment shown, aircraft **10** includes a pair of gas turbine engines. Each gas turbine engine **20** is mounted to aircraft **10** in an under-wing configuration. Each gas turbine engine **20** is capable of selectively generating propulsive thrust for the aircraft **10**. The gas turbine engine **20** may be configured to burn various forms of fuel including, but not limited to unless otherwise provided, jet fuel/aviation turbine fuel, and hydrogen fuel.

[0025] FIG. **2** depicts an exemplary gas turbine engine **20** defining an axial direction A (and centerline axis **22**) and a radial direction R. While the illustrated example shown is a high-bypass turbofan engine, the principles of the present disclosure are also applicable to other types of engines, such as low-bypass turbofans, turbojets, turboprops, unducted fan engines or open rotor engines, etc., as well as turbine engines having any number of compressor-turbine spools.

[0026] The gas turbine engine **20** includes a fan **24**, a low-pressure compressor **26** or "booster", a high-pressure compressor **28**, a combustion section or combustor **30**, a high-pressure turbine **32**, and a low-pressure turbine **34**, arranged in serial flow relationship. Collectively, the fan **24**, the low-pressure compressor **26**, and the low-pressure turbine **34** define a low-pressure system or low-pressure spool of the gas turbine engine **20**. Collectively, the high-pressure compressor **28** and the high-pressure turbine **32** define a high-pressure spool of the gas turbine engine **20**.

[0027] The high-pressure spool and combustor **30** may be referred to as a core engine **36** of the gas turbine engine **20**. The core engine **36** is at least partially enclosed within a core cowl **38**. The core cowl **38** may also at least partially enclose the low-pressure compressor **26** and the low-pressure turbine **34**. An engine casing **40** encases the core engine **36**. The engine casing **40** may include one or more of a compressor casing **42**, a compressor discharge casing **44**, a combustor casing **46**, and a turbine casing **48**. In exemplary embodiments, an under-cowl space **50** is defined between the engine casing **40** and the core cowl **38**. Various engine components may be positioned or stored within the under-cowl space **50**.

[0028] In particular configurations, a nacelle **52** surrounds at least a portion of the core engine **36**, the core cowl **38**, and the fan **24**. The nacelle **52** and the core cowl **38** form a bypass flow passage **54** therebetween. The nacelle **52** may be supported by one or more struts **56** that extend radially outward from an engine frame (not shown) to the nacelle **52**. A plurality of fuel injectors (one fuel injector **100** shown) is mounted to the engine casing **40**, more particularly, to the combustor casing **46**, and includes a portion that extends radially inward from the combustor casing **46** into the combustor **30**. A fuel supply system **58** is fluidly coupled to and in fluid communication with the fuel injector **100** to provide a flow of a fuel **60** such as a hydrocarbon fuel to the fuel injector **100**.

[0029] FIG. **3** provides a perspective view of an exemplary fuel injector **100** as may be incorporated into the gas turbine engine **20**, according to exemplary embodiments of the present disclosure. As shown in FIG. **3**, the fuel injector **100** may include a flange **102** adapted to be fixed and sealed to the combustor casing **46** (FIG. **2**). A hollow stem **104** extends inwardly with respect to radial direction R from the flange **102** and includes an inlet fairing **106** surrounding an annular pilot inlet (not shown) of the fuel injector **100**. In particular embodiments, the hollow stem **104** is integral and integrally formed with the inlet fairing **106**. In particular embodiments, the hollow stem **104** of the fuel injector **100** may be integral with (as illustrated herein) or fixed to the flange **102** (such as by brazing or welding).

[0030] As shown in FIG. **3**, the fuel injector **100** includes a valve housing **108** defining an outer surface **110**. The valve housing **108** extends radially outward from the flange **102** with respect to radial direction R. The valve housing **108** may be configured or formed to receive one or more fuel valves. For example, as shown in FIG. **3**, the valve housing **108** may be configured to receive a first fuel valve **112** and a second fuel valve **114**. In particular embodiments, the first fuel valve **112** may be configured as a pilot-fuel fuel valve and the second fuel valve **114** may be configured as a main-fuel fuel valve or vice versa. As shown in FIG. **3**, the hollow stem **104** and the inlet fairing **106** support a nozzle tip **116** including a nozzle body **118**. The first fuel valve **112** and the second fuel valve **114** are fluidly connected to and in fluid communication with fuel supply system **58** (FIG. **2**) via a first fuel inlet **120** of the first fuel valve **112** and a second fuel inlet **122** of the second fuel valve **114**. The nozzle body **118** is illustrated as having a dual-orifice pilot fuel and primary fuel injector tip configured to deliver both a flow of fuel for start-up operations and a primary flow of fuel for various ground, and in-flight operating conditions.

[0031] Referring now to FIG. **2**, in operation, fan **24** draws a first portion of air **62** into the bypass flow passage **54**. The first portion of air **62** is routed through the bypass flow passage **54** and out a bypass exhaust outlet **64** to provide primary thrust for the gas turbine engine **20**. A second portion of air **66** from fan **24** is drawn or routed into an inlet **68** of the low-pressure compressor **26** and is pressurized. The second portion of air **66** is further pressurized as it flows from the low-pressure compressor **26** and through the high-pressure compressor **28** to provide a high-pressure air **70** to a compressor discharge plenum **72** at least partially defined by the engine casing **40**.

[0032] The high-pressure air **70** flows from the compressor discharge plenum **72** into the combustor **30** where it is mixed with fuel **60** via fuel injector **100** and ignited, thereby generating combustion gases **74**. Work is extracted from the combustion gases **74** by the high-pressure turbine **32** which drives the high-pressure compressor **28** via a high-pressure shaft **76**. Combustion gases **74** then flow into the low-pressure turbine **34**, which drives the fan **24** and the low-pressure compressor **26** via a low-pressure shaft **78**.

[0033] As used herein, the gas turbine engine **20** is considered to be “operating” when fuel is being supplied to the fuel injectors **100**, burned in the combustor **30**, and the resulting combustion gases **74** are driving rotation of at least the core engine **36**. As used herein, gas turbine engine **20** is considered to be “shut down” when the fuel supply is shut off and is not being supplied to the combustor **30**. It will be understood that “operating” encompasses numerous operating conditions having varying rotor or shaft speeds and varying thrust and/or power outputs. It is also to be understood that parts of the engine might still be moving/slowing down, even though the engine is considered to be “shut down”.

[0034] During normal operations, temperatures of gas turbine engine components are maintained within allowable limits by a plurality of cooling systems that transfer heat from the various components to one or more heat sinks. When the engine is shutdown, most cooling systems no longer operate. Residual heat in certain engine components (e.g. “soakback”) can increase the temperature of other engine components beyond allowable limits. A particular concern is the formation of carbon (or “coke”) deposits in fuel carrying components, particularly a fuel injector, when a hydrocarbon fuel (liquid or gas) is exposed to high temperatures in the presence of oxygen.

[0035] FIG. **4** provides an enlarged schematic view of a portion of the gas turbine engine **20** including a portion of the core engine **36** as shown in FIG. **2**, and a fuel injector cooling system **200** according to exemplary embodiments of the present disclosure. The fuel injector cooling system **200** generally includes, in serial flow order, a blower **202**, a duct **204** fluidly connected to the blower **202** and at least partially defining a flow passage **206** downstream from the blower **202**, a valve **208**, and a cooling jacket **210** including an inner surface **212** and a cooling air inlet **214** in fluid communication with the flow passage **206**. The fuel injector cooling system **200** may be at least partially or fully enclosed within the under-cowl space **50** formed between the engine casing **40** and the core cowl **38**.

[0036] Duct **204** may be formed from one or more pipes, conduits, or the like, coupled together to, at least partially, define flow passage **206**. In particular embodiments, valve **208** may be a passive valve such as a butterfly valve. In other embodiments, valve **208** may be an actuator-controlled valve, such as but not limited to a butterfly valve, which is moveable between a fully open and a fully closed position via one or more of electrical, hydraulic, or pneumatic inputs provided by a controller **80**. Controller **80** may be a standalone controller, a gas turbine engine controller (e.g., a full authority digital engine control, or FADEC, controller), an aircraft controller, supervisory controller for a propulsion system, a combination thereof, etc.

[0037] Blower **202** includes a blower inlet **216** and a blower outlet **218**. In exemplary embodiments, blower inlet **216** may be in fluid communication with the under-cowl space **50** defined between the core cowl **38** and the engine casing **40**. In addition, or in the alternative, the blower inlet **216** may be in fluid communication with the low-pressure compressor **26**, the high-pressure compressor **28**, or with ambient air (A) from outside of the under-cowl space **50**, such as but not limited to the bypass flow passage **54**, via various blower inlet ducts **220** as shown in dashed lines. In an exemplary embodiment, blower **202** is electrically powered.

[0038] In exemplary embodiments, duct **204** includes an upstream end **222** that is fluidly coupled to and in fluid communication with the blower outlet **218**, and a downstream end **224** that is fluidly coupled to and in fluid communication with the valve **208** via a valve inlet **226**. In particular embodiments, a second duct **228** is fluidly coupled to and in fluid communication with a valve outlet **230** of valve **208**. The second duct **228** may be formed from one or more pipes, conduits, or the like, coupled together to further define the flow passage **206** downstream from the valve outlet **230**.

[0039] The second duct **228** may be fluidly connected to and in fluid communication with a cooling air plenum **82** at least partially defined within the engine casing **40** and disposed downstream from valve outlet **230**. The cooling air plenum **82** may be in fluid communication with the combustor **30** for providing cooling air to portions of the fuel injector **100** disposed within the engine casing **40** and the combustor **30**, particularly the hollow stem **104**, the inlet fairing **106**, the nozzle tip **116**, and the fuel nozzle body **118**, during shut down of the gas turbine engine **20** to prevent or reduce the formation of coke within those portions of the fuel injector **100**. In exemplary embodiments, a check valve **232** may be positioned within the second duct **228** upstream from the cooling air plenum **82**. Check valve **232** may be configured to prevent backflow of the high-pressure air **70** into the second duct **228** when the gas turbine engine **20** is operating.

[0040] In exemplary embodiments, the cooling air inlet **214** of the cooling jacket **210** is in fluid communication with the flow passage **206** via an air extraction port **234** disposed along and in fluid communication with the duct **204**. In particular embodiments, the air extraction port **234** is disposed and fluidly coupled to the flow passage **206** between the blower outlet **218** and the valve inlet **226**.

[0041] FIG. **5** provides a perspective aft facing view of the fuel injector **100** including the cooling jacket **210** surrounding at least a portion of the valve housing **108** according to exemplary embodiments of the present disclosure. FIG. **6** provides a cutaway view of a top portion of the fuel injector **100** including the cooling jacket **210**, according to exemplary embodiments of the present disclosure. As shown in FIGS. **6** and **7** collectively, the cooling jacket **210** is formed or shaped to extend at least partially around at least a portion of the valve housing **108** of the fuel injector **100**.

[0042] As illustrated in FIG. **6**, the inner surface **212** of the cooling jacket **210** is offset from the outer surface **110** of the valve housing **108** to form a cooling flow passage **236** between the inner surface **212** of the cooling jacket **210** and the outer surface **110** of the valve housing **108**. The cooling jacket **210** defines at least one cooling flow outlet **238**(*a*-*c*). For example, in exemplary embodiments, a first cooling flow outlet **238**(*a*) is formed proximate to the flange **102**. A second cooling flow outlet **238**(*b*) may be formed proximate to the first fuel inlet **120** of the first fuel valve **112**. In addition, or in the alternative, a third cooling flow outlet **238**(*c*) may be formed proximate to the second fuel inlet **122** of the second fuel valve **114**. The first fuel inlet **120** and the second fuel inlet **122** are fluidly connected to and in fluid communication with the fuel supply system **58** (FIG. **2**).

[0043] FIG. **7** provides an exploded perspective view of the cooling jacket **210** according to exemplary embodiments of the present disclosure. FIG. **8** provides an enlarged view of a portion of the cooling jacket **210** and the valve housing **108** at a respective cooling flow outlet **238** according to exemplary embodiments of the present disclosure. As shown in FIG. **7**, the cooling jacket may be formed from two or more shell bodies. For example, in the embodiment shown in FIG. **7**, the cooling jacket includes a first shell body **240** and a second shell body **242**. The first shell body **240** and the second shell body **242** may be clamped together in a “clam shell” or other configuration to seal the mating shell bodies. It is to be appreciated that although the cooling air inlet **214** is shown on the first shell body **240**, it may, in the alternative, be formed on the second shell body **242**. One or more mechanical fasteners **244** such as one or more clamps may be used to mechanically fix the first shell body **240** to the second shell body **242**. In addition, or in the alternative, the mechanical fastener **244** may include a bolt (not shown) to secure the first shell body **240** to the second shell body **242**. In other embodiments, the first shell body **240** and the second shell body **242** may be welded or otherwise joined together. It is to be appreciated that, in alternate embodiments, the cooling jacket **210**, or more particularly, the cooling flow passage **236**, may be integrally formed, such as via additive manufacturing or other known manufacturing processes, with the valve housing **108**.

[0044] In exemplary embodiments, as shown in FIG. **8**, a plurality of spacers **246** are disposed between the outer surface **110** of the valve housing **108** and the cooling jacket **210** to provide a gap **248** between the outer surface **110** and the inner surface **212**, thus at least partially forming the cooling flow passage **236** therebetween. As shown in FIGS. **7** and **8** collectively, the plurality of spacers **246** may be at least partially or fully defined by the first shell body **240** and the second shell body **242** of the cooling jacket **210**.

[0045] FIG. **9** provides a perspective view of a portion of the fuel injector cooling system **200** according to an exemplary embodiment of the present disclosure. As shown in FIG. **9**, the fuel injector cooling system **200** may include a plurality of cooling jackets **210** with each cooling jacket **210** surrounding a respective valve housing **108** of a respective fuel injector **100** (not shown). Each cooling jacket **210** is fluidly coupled to and in fluid communication with a ring manifold **250**. Ring manifold **250** is fluidly connected to and in fluid communication with the flow passage **206** via one or more fluid couplings such as pipes or conduits **252**. Ring manifold **250** extends circumferentially about the centerline axis **22** and around the engine casing **40** (FIG. **4**). In exemplary embodiments, as shown in FIG. **9**, ring manifold **250** is in fluid communication with the flow passage **206** via the air extraction port **234** downstream from the blower outlet **218** and upstream from the valve **208**.

[0046] Referring to FIGS. **4-9** collectively, in operation, blower **202** of the fuel injector cooling system **200** is switched on when the gas turbine engine **20** is shutdown, or in anticipation of a shutdown, or in certain instances where a cooling system normally used to cool the fuel injector(s) **100** may be compromised. Blower **202** draws air from one or more of the under-cowl space **50**, the high-pressure compressor **28**, the low-pressure compressor **26** and the bypass flow passage **54**, into the flow passage **206**. At least a portion of the air is routed to the cooling air inlet **214**, of the cooling jacket **210** via the air extraction port **234**, into the cooling flow passage **236** of the cooling jacket **210**. In the alternative, at least a portion of the air is routed to the ring manifold **250** to be distributed to the respective cooling air inlets **214** of each of the plurality of cooling jackets **210**. The air flows through each respective cooling flow passage **236** of each respective cooling jacket **210**, across the outer surface **110** of each respective valve housing **108**, thus cooling the respective valve housing **108** and reducing or preventing the potential formation of coke therein. In particular embodiments, a second portion of the air may be routed from the flow passage into the cooling air plenum **82** where it may be directed across the portion of the fuel injector **100** that is disposed within the engine casing **40** and within the combustor **30**.

[0047] Further aspects are provided by the subject matter of the following clauses:

[0048] A fuel injector cooling system, comprising: a blower including a blower inlet and a blower outlet; a valve including a valve inlet and a valve outlet; a duct defining a flow passage between the blower outlet and the valve inlet; a fuel injector including a flange and a valve housing extending radially outward from the flange, wherein the valve housing defines an outer surface of the fuel injector; and a cooling jacket defining an inner surface and a cooling air inlet in fluid communication with the flow passage of the duct, wherein the cooling jacket at least partially surrounds the valve housing to define a cooling flow passage between the inner surface of the cooling jacket and the outer surface of the valve housing, wherein the cooling flow passage is in fluid communication with the flow passage.

[0049] The fuel injector cooling system of the preceding or any following clause, wherein the cooling jacket comprises a first shell body and a second shell body.

[0050] The fuel injector cooling system of any preceding or following clause, further comprising a mechanical fastener, wherein the mechanical fastener couples the first shell body to the second shell body.
[0051] The fuel injector cooling system of any preceding or following clause, wherein the cooling jacket defines a spacer extending between the inner surface of the cooling jacket and the outer surface of the valve housing.
[0052] The fuel injector cooling system of any preceding or following clause, further comprising a ring manifold in fluid communication with the flow passage and the cooling air inlet of the cooling jacket.
[0053] The fuel injector cooling system of any preceding or following clause, further comprising an air extraction port disposed along the duct between the blower outlet and the valve inlet, wherein the cooling air inlet is in fluid communication with the flow passage via the air extraction port.
[0054] The fuel injector cooling system of any preceding or following clause, wherein the fuel injector cooling system is integrated into a gas turbine engine, the fuel injector cooling system further comprising a check valve disposed downstream from the valve outlet, wherein the check valve is in fluid communication with a cooling air plenum of a gas turbine engine.
[0055] The fuel injector cooling system of any preceding or following clause, wherein fuel injector further comprises a fuel nozzle body, wherein the fuel nozzle body is in fluid communication with the cooling air plenum.
[0056] The fuel injector cooling system of any preceding or following clause, wherein the gas turbine engine includes an under-cowl space, wherein the blower inlet is in fluid communication with the under-cowl space.
[0057] The fuel injector cooling system of any preceding or following clause, wherein the valve housing of the fuel injector is configured to receive a first fuel valve and a second fuel valve.
[0058] A gas turbine engine, comprising: a core engine including, in serial flow order, a high-pressure compressor, a compressor discharge plenum, a combustor surrounded by an engine casing, and a plurality of fuel injectors, each respective fuel injector of the plurality of fuel injectors having a nozzle body disposed inside of the engine casing within the combustor and a valve housing disposed outside of the engine casing; and a fuel injector cooling system, comprising: a blower including a blower inlet and a blower outlet; a valve including a valve inlet and a valve outlet; a duct defining a flow passage between the blower outlet and the valve inlet; and a plurality of cooling jackets, each respective cooling jacket of the plurality of cooling jackets including an inner surface and a cooling air inlet in fluid communication with the flow passage, wherein each respective cooling jacket defines a cooling flow passage between the inner surface of the respective cooling jacket and the outer surface of the valve housing of the respective fuel injector of the plurality of fuel injectors.
[0059] The gas turbine engine of the preceding or any following clause, wherein each respective cooling jacket of the plurality of cooling jackets includes at least one spacer extending between the inner surface of the respective cooling jacket and the outer surface of the respective fuel injector.
[0060] The gas turbine engine of any preceding or any following clause, wherein at least one respective cooling jacket of the plurality of cooling jackets comprises a first shell body and a second shell body.
[0061] The gas turbine engine of any preceding or any following clause, further comprising a core cowl surrounding at least a portion of the engine casing, wherein an under-cowl space is defined between the engine casing and the core cowl, wherein the fuel injector cooling system is disposed within the under-cowl space.
[0062] The gas turbine engine of any preceding or any following clause, wherein the blower inlet is in fluid communication with the under-cowl space.
[0063] The gas turbine engine of any preceding or any following clause, further comprising a ring

manifold extending circumferentially about the engine casing, wherein the ring manifold is in fluid communication with the flow passage and with the cooling air inlet of each respective cooling jacket of the plurality of cooling jackets.

[0064] The gas turbine engine of any preceding or any following clause, further comprising an air extraction port disposed along the duct between the blower outlet and the valve inlet, wherein the cooling air inlet of each respective cooling jacket of the plurality of cooling jackets is in fluid communication with the flow passage via the air extraction port.

[0065] The gas turbine engine of any preceding or any following clause, wherein the valve housing of at least one of the respective fuel injectors of the plurality of fuel injectors is configured to receive a first fuel valve and a second fuel valve.

[0066] The gas turbine engine of any preceding or any following clause, further comprising a cooling air plenum defined within the engine casing, wherein the fuel injector cooling system further comprises a check valve disposed downstream from the valve outlet, wherein the check valve is in fluid communication with the cooling air plenum.

[0067] The gas turbine engine of any preceding or any following clause, wherein each respective fuel injector of the plurality of fuel injectors includes a nozzle body in fluid communication with the cooling air plenum.

[0068] This written description uses examples to disclose the present disclosure, including the best mode, and also to enable any person skilled in the art to practice the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A fuel injector cooling system, comprising: a blower including a blower inlet and a blower outlet; a valve including a valve inlet and a valve outlet; a duct defining a flow passage between the blower outlet and the valve inlet; a fuel injector including a flange and a valve housing extending radially outward from the flange, wherein the valve housing defines an outer surface of the fuel injector; and a cooling jacket defining an inner surface and a cooling air inlet in fluid communication with the flow passage of the duct, wherein the cooling jacket comprises a first shell body and a second shell body, wherein the cooling jacket at least partially surrounds the valve housing to define a cooling flow passage between the inner surface of the cooling jacket and the outer surface of the valve housing, wherein the cooling flow passage is in fluid communication with the flow passage, and wherein the cooling flow passage defines an inlet at a portion of the valve housing radially outward from the flange and an outlet at a portion of the valve housing proximate to the flange.

2. The fuel injector cooling system of claim 1, wherein the first shell body includes a first plurality of spacers and the second shell body includes a second plurality of spacers, wherein the first plurality of spacers and the second plurality of spacers provide a gap between the outer surface of the valve housing and the inner surface of the cooling jacket, wherein the gap forms the cooling flow passage, and wherein the plurality of spacers project radially inward to contact the valve housing.

3. The fuel injector cooling system of claim 2, further comprising a mechanical fastener that couples the first shell body to the second shell body.

4. (canceled)

5. The fuel injector cooling system of claim 1, further comprising a ring manifold in fluid

communication with the flow passage and the cooling air inlet of the cooling jacket.
6. The fuel injector cooling system of claim 1, further comprising an air extraction port disposed along the duct between the blower outlet and the valve inlet, wherein the cooling air inlet is in fluid communication with the flow passage via the air extraction port.
7. The fuel injector cooling system of claim 1, wherein the fuel injector cooling system is integrated into a gas turbine engine, the fuel injector cooling system further comprising a check valve disposed downstream from the valve outlet, wherein the check valve is in fluid communication with a cooling air plenum of the gas turbine engine.
8. The fuel injector cooling system of claim 7, wherein the fuel injector further comprises a fuel nozzle body in fluid communication with the cooling air plenum.
9. The fuel injector cooling system of claim 7, wherein the gas turbine engine includes an under-cowl space in fluid communication with the blower inlet.
10. The fuel injector cooling system of claim 1, wherein the valve housing of the fuel injector is configured to receive a first fuel valve and a second fuel valve.
11-20. (canceled)
21. A gas turbine engine, comprising: a core engine including, in serial flow order, a high-pressure compressor, a compressor discharge plenum, a combustor surrounded by an engine casing, and a plurality of fuel injectors, each respective fuel injector of the plurality of fuel injectors having a nozzle body disposed inside of the engine casing within the combustor and a valve housing disposed outside of the engine casing; and the fuel injector cooling system of claim 1.